

Task 1

Deadline/Presentation 28.05
(Upload at Felix **before** the lecture,
use email if this does not work)

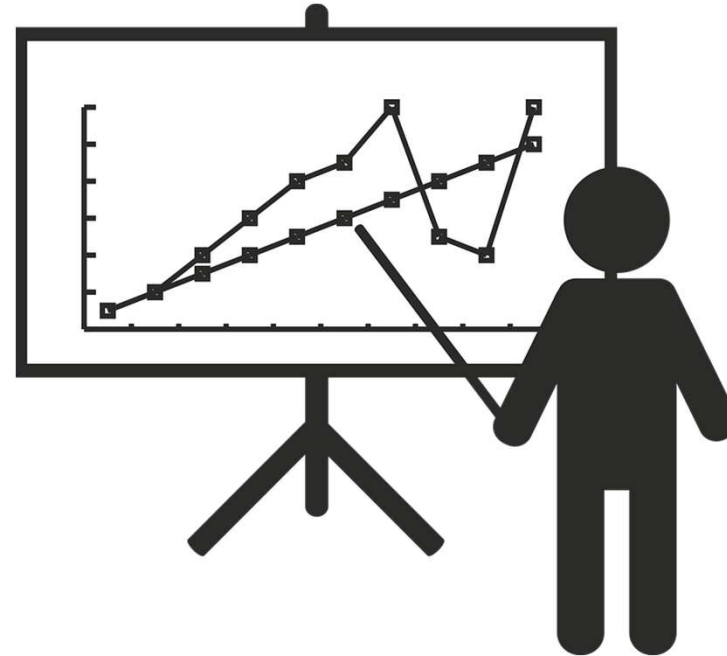
What if you did the SbA already

- Send me an Email with the EXACT subject line
“IBS5: Data Science sbA from earlier term” until April 23rd
- Include in the mail
 - Your name
 - Your student id
 - The term when you did the sbA
 - Information about the group you did it with
(Group number and or names of group members)
- Make sure that you get a confirmation from me about having verified your earlier submission
- Send me reminder with the same subject line if you do not hear back from me by the 1st of May.

Skills of a Data Scientist



Generating
Insights from Data



Communicating
Insights

Technology to use

- The task is to be implemented with Python and optionally other tools (e.g. for additional visualization).
- The majority of the work is to be implemented with Python.

Presentation requirements

- **Group size max 6, min 3 (→ register with Felix) (5 is the ideal size). (Exceptions need approval)**
 - Individual evaluation possible: → register in advance and clearly mark contribution of each group member
- **Felix is not used to find the groups (just to reflect them)**
- **Contact me via email if you do not find a group**
- **Send reminders if you do not get a response for an email**

Grading of Presentation

- **Timing!**
- Completeness of the results
- Reflection on results
 - What value can we derive from your insights?
 - Why was the analysis done in this particular way?
 - What could be improved (if you had more time and resources)?
- Is the presentation well tailored to the target group (easy to comprehend)
 - Target group is CEO/CTO of „made up“ company that hired your team
- How you present
- Discussion

Duration of Presentations

- Presentation times
 - 3 persons: 12 min
 - 4 persons: 15 min
 - 5 persons: 19 min
 - 6 persons: 23 min
- **Timing is strict! You will not have more time. If time runs out, you need to stop. No extra time.**
- (Somewhat odd numbers due to time constraints.)
- Groups will start in random order.

Background/Motivation for tasks

- Made-up manufacturer SmartBuild
 - SmartBuild builds a final product from raw material in production lines
 - The properties like width, height, ionizationclass, FluxCompensation, pressure, karma, modulation of the raw material are measured on input.
 - Settings of the machine are logged
 - Properties of the resulting end product are also measured (other attributes in the data)
 - SmartBuild wants to check if the properties can be predicted at the input to optimize the production (e.g. avoid rejects/defects, increase quality).

Background/Motivation for tasks

- You are a consulting team with expertise in Data Science
- You have been hired by SmartBuild to explore the potential of Data Science technologies for the company's manufacturing operations



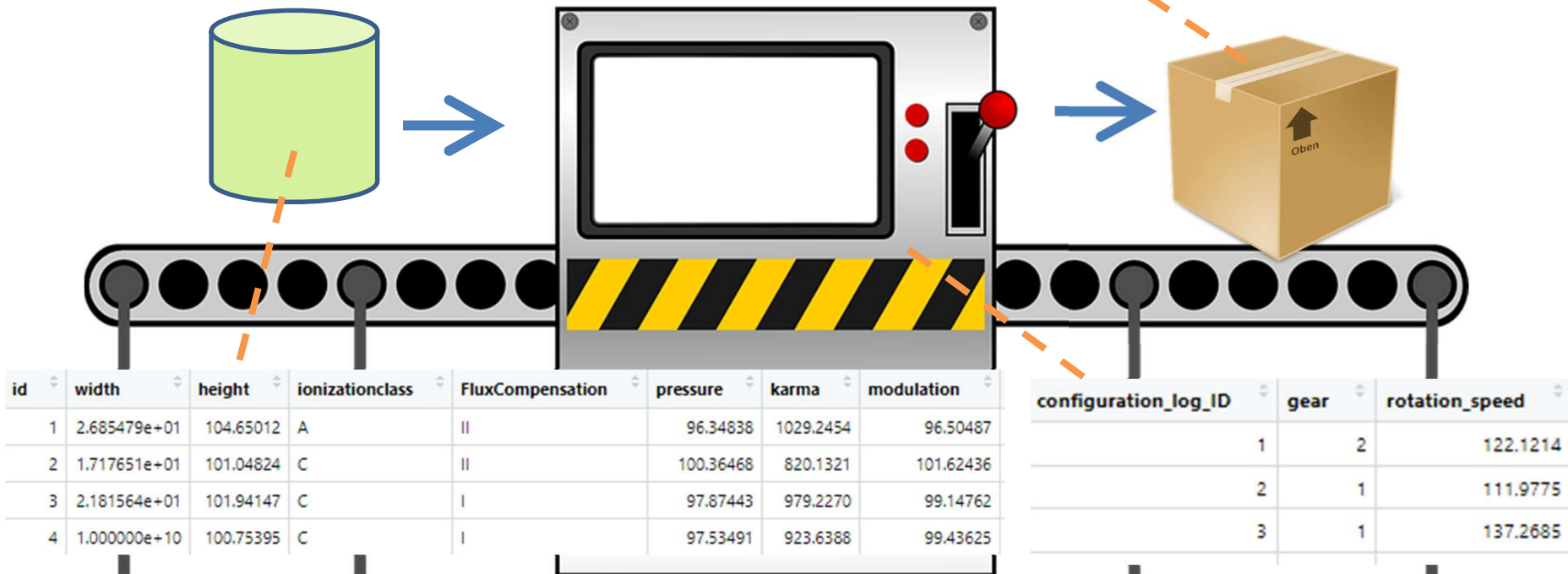
Data for the Task

- The data can be found in Felix's files Machine_Settings_Log_01.csv and Production_Log_01.csv
- Particularly interesting is the prediction of errors before the machine is used.
- A faulty product costs 150 EUR (i.e. use of the machine + material costs)
- The raw material for one product costs 10 EUR. (Discarding/sorting it out means a loss of 10 EUR.)

SmartBuild Data Capturing

Properties of the final product after the machine is used

weight_in_kg	weight_in_g	error	error_type	Quality	reflectionScore	distortion	nicesness	smartness	multideminsionality	roughness
1344159.8	1344159804	yes	severe	93.56282	759.3351	50.47403	99.55477	21057.41	no	120.5376
442485.8	442485753	no	None	98.61264	701.9900	51.14783	57.93089	17408.30	no	113.3320
759766.1	759766111	no	None	99.02346	703.9903	44.05320	77.53556	17157.99	yes	137.4725



Measured before the machine is used

Log of the respective machine settings

Assessment of the Technical Solution

- Evaluation criteria
 - Overall quality of the solution
 - Depth of experiments/tests
 - Sound evaluation/interpretation of the results
 - Reasonable design of the models (features, parameters)
 - Solution for pre-processing of data (if available)

Assessment of the Presentation

- Evaluation criteria
 - Timing
 - Targeted to the Made-up target group CEO+CTO of the Made-up manufacturing company to be consulted.
 - Note: The company wants to understand your results and the technical foundation of your solution.
 - Potentially they would like to integrate your solution into their corporate IT

Tasks

- Address Q1,Q2
- Create predictive models for further product characteristics so that you have at least 2 analyses per person (different models, type of model, or feature sets)
- “One analysis” is deliberately a vague concept. Fill the presentation time with interesting insights for the “customer”
- Explain the concepts you used for data analysis (for the target group)
- Evaluate and review your results
- **Note: If applicable, you can also show that a good prediction (using the models we have covered) is not possible.**

Q1

- Create a predictive model for the attribute "weight_in_kg".

Q2

- Create a predictive model for the "error_type" attribute

Q2 Alternative

- Create a prediction model for the "error" attribute

Use of LLMs (e.g. ChatGPT)

- For **this** course the following applies until further notice:
 - LLMs are treated like existing programming tools or help pages. You do not have to **denote the use in this class**.
 - *You must be able to explain your solution in detail (i.e. you must understand what you are doing). → Presentation discussion*
 - *Note: This policy does not apply to other classes*